CSA1701-ARTIFICIAL INTELLIGENCE LAB EXPERIMENTS

Program 7 : Python Program to Implement BFS

Aim

To develop a Python program that traverses a graph using the Breadth First Search (BFS) algorithm.

Algorithm

- Represent the graph as an adjacency list using a dictionary.
- Read the starting node and place it in the queue.
- Remove the first node from the queue.
- If the node is not visited, print it and add it to the visited list.
- Add all neighbours of the node to the end of the queue.
- Repeat until the queue becomes empty.

Program

```
graph = {     'A':
['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': [] } start = input("Enter
starting node: ")

queue = [start]
visited = []

while queue:     node =
queue.pop(0)

    if node not in visited:
print(node, end=" ")
visited.append(node)
queue.extend(graph[node])
```

Output

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============== RESTART: C:/Users/shaik/OneDrive/Documents/7.py ==============
Enter starting node: A
A B C D E F
>>>
```

RESULT

Thus, the BFS traversal was implemented successfully using Python.